

The whole story in six lines

1. Big projects almost always cost more and take longer than the number the board approved. In oil and gas: 64% over budget, 73% late, 59% average cost growth, and the Middle East is the worst region measured (89% over cost). *(EY, 365 projects over \$1bn)*
2. It isn't a software shortage. It's that the facts that would have saved the project are spread across dozens of systems, dozens of companies and millions of documents nobody can read fast enough.
3. Reading is now cheap. That is the only thing that changed, and it changes which problems are solvable.
4. So we sell three reading machines, in the order the owner can actually buy them.
5. Each one pays for itself out of a single avoided loss on a single project.
6. None of them asks the owner to change how they work. That's the design constraint, not a nice-to-have.

What goes wrong, in money

WHAT HAPPENS	HOW BIG	SOURCE
Contractors ask for roughly twice what they eventually settle for	~2× gap between submitted and settled	IPA via National Academies / Long International
Money in dispute on affected projects	33.4% of the contract budget	HKA CRUX (2,200+ projects)
One average dispute	\$60.1M and about 12.5 months	Arcadis 2025
Redoing work that was built wrong	~5% of project cost (top 10% of projects: 12.4%)	CII IR-153
Where those errors come from	70-80% of cost deviations trace back to design, not building	industry analyses cited in the report
Cost of catching an error late	10× more in construction, 100× more after handover	CII 1-10-100
Overrun after the board has already approved the budget	23% on average, even post-sanction	EY

Plain reading: the money leaks in three places. During the build, through claims. In the design office, through errors that get built. And at the moment of approval, through a number that was too hopeful.

Our three products sit exactly on those three leaks.

Why decades of software didn't fix it

- Every project is a one-off. Teams disband, lessons evaporate, and the next project starts from zero.
- The contract makes truth adversarial. Fixed-price deals reward claiming and reward hiding bad news.
- Low estimates win approval, so estimates come in low. That's not a mistake, it's an incentive.
- Reporting is monthly. Any problem is four to eight weeks old before a leader sees it.
- The decisive evidence is prose: letters, minutes, daily reports, drawing comments. Old software could only handle the tidy 20% of data in databases, and the messy 80% was invisible.

The gap is not analysis, it's reading. That's the sentence the whole business rests on.

The owner's lifecycle, and where the money is really decided

IDEA (screen)	OPTIONS (pick one)	DESIGN (FEED)	APPROVE (the vote)	BUILD (3-6 years)	START-UP (handover)	OPERATE (20-40 yrs)
 cheap to kill	 cheapest moment to change your mind	 where cost is actually set	 the number everything is measured against	 where the overrun becomes visible	 where delay costs most (money spent, no revenue)	 where bad handover data costs for decades

Two facts that drive our whole product order:

1. Cost is **decided** in design and at the approval vote. It only **shows up** years later, during the build.
2. Almost every AI product in this market works during the build, for the contractor. The owner's seat, the design stage and the approval gate are open ground.

The three products on that map

IDEA	OPTIONS	DESIGN	APPROVE	BUILD	START-UP	OPERATE
		[PRODUCT 2]	[PRODUCT 3]	[PRODUCT 1]		
		Design feasibility	FID red team	Claims defence		
		check every revision	pressure-test	& recovery		
			the number	always on		
		~ ~ later opportunities ~ ~				
concept	option	drawing chase	gate challenge	cost forecast truth	handover	asset data
red-team	compare	(deliverables)	book	schedule truth	complete-	feeds the
				interface gaps	ness	next vote

- **Product 1 — Claims defence and recovery.** Ships first. Runs during the build, on the record the owner already controls.
- **Product 2 — Design feasibility.** Ships alongside or just after. Runs during design, on the owner's own drawings.
- **Product 3 — Approval red team (FID).** Switched on after 1 and 2 have earned trust, because it needs the owner's history and it delivers an unwelcome message at the highest-stakes moment.

The greyed items are the roadmap. Each one reuses the same reading layer, so the second product costs a fraction of the first.

Product 1: what it actually does

For the owner's contracts and commercial manager on a live build.

It reads the contract, the letters, the schedule updates and the site reports, every day, and keeps a running list of money moves worth making:

- "This \$40M claim rests on six events. On three of them the contractor missed its own notice deadline. Here's the rebuttal, with the letters cited."
- "You can charge \$2.1M of late-delivery damages this month. Here's the calculation and the evidence."
- "This letter, dated 14 May, signals a claim that isn't in your risk register yet."

Every item shows the money at stake, how confident the agent is, and the documents behind it. The manager approves, edits or rejects. The agent writes the letter. A person sends it.

Two things make it credible: it never sends anything itself, and every line clicks through to a page in a real document.

Product 1: what it's worth, and what it costs you

Value (money). On a \$5bn project, our conservative model:

STEP	NUMBER	WHERE IT COMES FROM
Claims submitted over the build	~\$400M (8% of project value)	our assumption, to validate with the owner's own
Typically settled at about half	~\$200M paid	IPA: submitted vs settled gap "as much as half"
Sharper, evidenced defence on 10% of that	\$20M kept	our model
Late-delivery damages and back-charges pursued properly (from ~40% to ~70% of what's	~\$9M recovered	our model, using a \$30M entitlement pool

Product 2: what it actually does

For the owner's design and engineering manager during design.

Every time a drawing set is revised, the agent checks three things in minutes and hands back a cited pass or fail list:

- **Is it legal?** Checked against the building code that applies (Saudi Building Code, Dubai Building Code).
- **Is it affordable?** Checked against the budget ceiling for that part of the works.
- **Can it be built?** Clashes, access, and equipment with long delivery times that the sequence assumes will be there.

It's a spell-check for feasibility. It catches the error while it costs thousands, not the tens of millions it becomes once it's in the ground.

Strongest demand signal in the market: Dubai Municipality is building a system to issue permits automatically by reading submitted drawings against the Dubai Building Code. When the regulator automates the check, every owner has to arrive pre-compliant.

Product 2: what it's worth, and what it costs you

Value (money). On a \$5bn project:

STEP	NUMBER	SOURCE
Rework at ~5% of project cost	~\$250M	CII IR-153
Share caused by design (~28%)	~\$70M	industry analyses in the report
Catch half of it in design	~\$35M avoided per project	our model on sourced inputs
On a \$20bn project, the design-caused pool	~\$280M	same method
Review time	6 weeks down to 3 days on Saudi Building Code review	WhiteHelmet customer testimonial, 2026 (self-reported)
Independent return modelling	4:1 to 30:1 cost avoidance; a query costs \$200–500 at drawing stage vs \$2,000–8,000 in the field	Mirage Metrics, 2026

Likelihood of delivering: high on a narrow scope, medium if we promise everything. Reading drawings and code is now doable, and two Gulf vendors already sell a version of it. The honest risk is coverage: we start with one code set and the disciplines that cause the most rework, and we say so.

Effort from the persona: lowest of the three. It hooks into the design file drop that already happens. One-time job: confirm which code edition and which budget ceilings apply. After that the designer's habit doesn't change at all.

Sacrifice from the persona: professional pride, mostly. A machine is marking an engineer's work inside the engineer's own workflow. Our

Product 3: what it does, and why it waits

For the project director, the lead estimator and the investment committee, at the approval vote.

Before the board commits, the agent pressure-tests the number: it compares this project against how comparable past projects actually turned out, audits the design package for the gaps that predict overruns, and writes the challenge book for the gate meeting.

- "This project looks like a group that overran 60%. Here are its nine weakest scope items."
- "Your productivity assumption is above what comparable Gulf projects actually achieved."

Why it waits: it needs the owner's own project history, and it tells the most senior people in the room that their number is wrong. That's a sale you make once you're already trusted and already inside the data. Products 1 and 2 buy that position.

The prize is the biggest, though: EY finds a 23% average overrun **after** approval. On a \$5bn approval, that's \$1.15bn of exposure. Taking a fifth of it off the table is \$230M.

The three side by side

	1. CLAIMS DEFENCE & RECOVERY	2. DESIGN FEASIBILITY	3. APPROVAL RED TEAM
Persona	Contracts / commercial manager	Design / engineering manager	Project director, estimator, investment committee
Stage	Build	Design	The approval vote
Value per \$5bn project	~\$30M	~\$35M	~\$230M of exposure addressed
Likelihood of delivering	High	High on a narrow scope	Medium
Effort from the persona	Low: read-only files, 20 min/day	Lowest: hooks existing file drop	High: hand over portfolio history
Sacrifice	Trusting software with privileged commercial material	Being marked by a machine	Political: an unwelcome message at the biggest vote
Data needed	Contract, letters, schedule, site reports	Drawings, code set, budget ceilings	Historical outcomes across the portfolio
Verdict	Ship first	Ship in parallel	Earn the right, then ship

The rule behind the order: value divided by (effort + sacrifice), with delivery confidence as the multiplier. Product 3 wins on value alone and loses on everything else, which is exactly why it's third and not never.

The adoption promise (this is the real product decision)

The fastest way to lose this market is to ask a project team to change how it works mid-project. So we commit to six rules:

1. **Runs on files they already have.** Read-only. No migration, no new system of record.
2. **No new habit.** It arrives where they already look (email or Teams), and reviews happen on one screen.
3. **The agent never sends.** It drafts. A person presses send. Always.
4. **Every claim is clickable.** No assertion without a citation to a page in a real document.
5. **Proof on history first.** We run on last year's closed files and show what it would have caught, before it touches a live project.
6. **Priced against one avoided loss,** and sized so a pilot sits inside one project team's own authority.

That last one matters commercially: ADNOC signed a \$340M three-year agentic-AI contract, but only after a proof of concept. The deal shape in this market is small paid pilot, then large multi-year contract.

Proof from people who live it

"The difference between the amount submitted for a claim versus the amount actually paid at settlement... is as much as half. The owner too must ensure profitability of his project by defending unsupportable claims by a 'low ball' contractor."

IPA via National Academies and Long International. *Product 1, stated from the owner's seat.*

"What most projects lose to poor claims management is not a dispute they should have won, but entitlement they never assembled the evidence to claim."

Kairos, claims advisory, 2026. *The failure mode named exactly.*

"A fully integrated digital system capable of issuing building permits automatically and without human intervention by reading submitted drawings and documents, verifying compliance with the Dubai Building Code."

Dubai Municipality, 2026. *Product 2's demand signal, from the regulator.*

"We reduced our compliance review cycle from 6 weeks to 3 days. The AI caught two fire environmental gaps that would have delayed our building permit by months."

WhiteHelmet customer, 2026 (self-reported). *Product 2, quantified in the Gulf.*

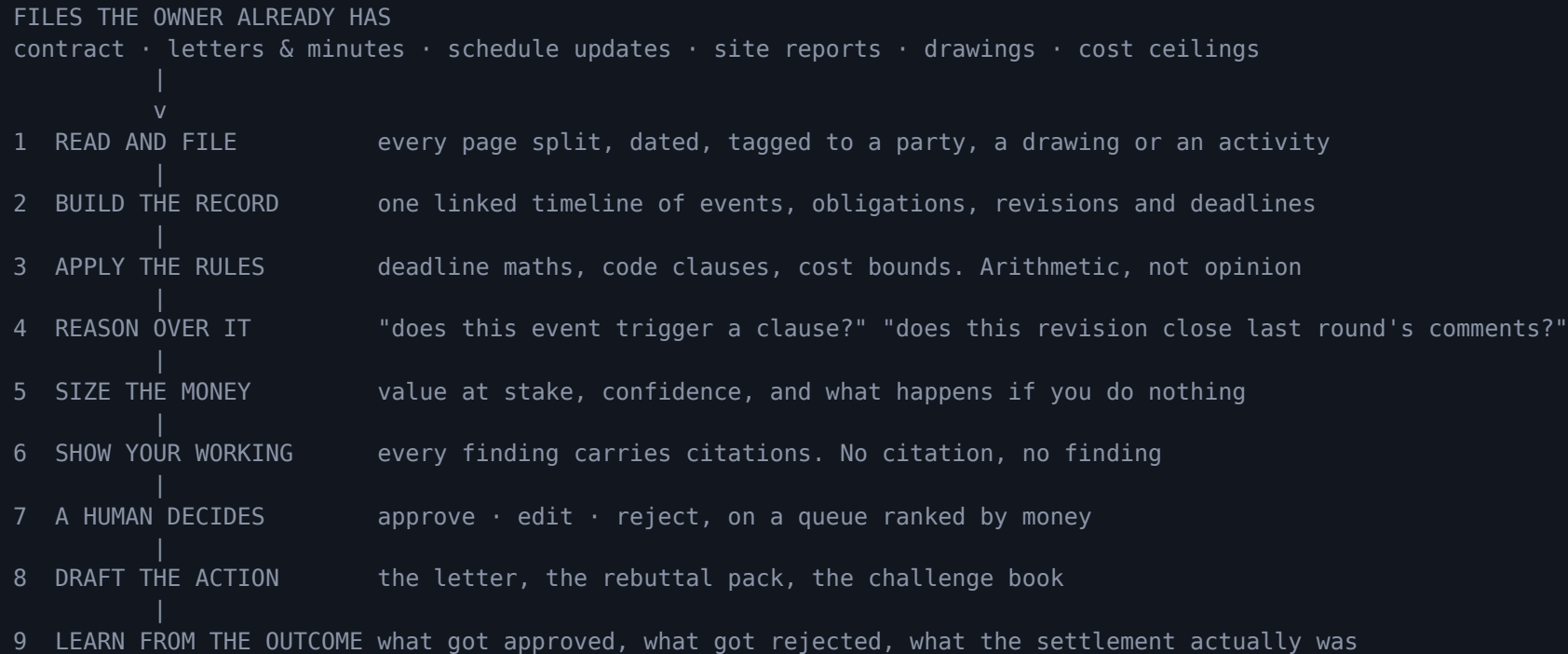
"More than half of its potential lost to underperformance... all shared a root cause: bias."

McKinsey on an owner's capital portfolio, 2025. *Product 3's reason to exist.*

"ADNOC is on a mission to become the world's most AI-enabled energy company." \$340M, three years, after a proof of concept.

Musabbeh Al Kaabi, ADNOC Upstream CEO, 2025. *The willingness to pay, and the price ceiling.*

How it works, in one picture



Two engineering commitments behind that picture:

- **The hard maths is deterministic, the soft reading is the model's job.** Deadlines, damages and cost bounds are computed in code so they're always reproducible. The model reads and proposes, it doesn't do the arithmetic that ends up in a letter.
- **Nothing ships without an eval.** Each skill has a scored test set built from closed projects: did it find the events a specialist found, did it invent anything, did it get the deadline right. Releases are gated on those scores.

How we'll know it's working

Business. Money kept or recovered per project (the number the pilot is judged on), pilot to contract conversion, share of project value captured in price.

Product. Items reviewed per week per manager, share of drafted actions actually sent, time from an event appearing in the record to a human deciding on it, and one honest counter-metric: how often a manager overrides the agent, and whether that falls.

AI engineering. Recall against a specialist's own findings on closed projects, invented-fact rate at zero tolerance for cited claims, deadline accuracy, and cost per document read.

Risks, and what we do about them

RISK	WHAT WE DO
Owner won't give a machine the privileged commercial record	Start on closed projects. In-region private cloud. Permissions that mirror the existing team. Full read audit.
Wrong finding damages a legal position	Draft only, never send. Everything cited. Deadline and damages maths in code, not model output.
Code coverage in Product 2 is thinner than promised	Narrow first release: one code edition, the disciplines that cause the most rework. Publish what's in scope.
Incumbents move (Trimble, Procore, and the AI vendors)	They're contractor-side and build-phase. We're owner-side across design, approval and build, on data they don't have.
Product 3 needs history the owner may not have organised	Bundle the history clean-up as the first paid step, and lean on public outcome data until it exists.
Buying gates in the Gulf (registration, local content, cyber certification, data residency)	Treat them as features on the roadmap from day one, not paperwork at the end.

First 90 days

- **Days 1-15.** Ten interviews: owner contracts managers, design managers, one estimator, one investment-committee member. Validate the \$400M claims-submission assumption against a real portfolio.
- **Days 16-45.** Product 1 on one closed project's archive. Success test: it finds what the commercial team found, plus at least one thing they missed, all cited.
- **Days 30-60.** Product 2 on one FEED revision history. Success test: it flags the errors that actually became field changes.
- **Days 60-90.** One paid pilot signed on a live project, with the money-kept metric agreed in writing before we start. Compliance work (hosting, cyber certification, local content) running in parallel.

Product 3 gets a retrospective demo only in this window: "here are the twelve flags we'd have raised at your last approval." No sale attached yet.

Where this goes

Same reading layer, more skills, in the order the owner will accept them:

- **During the build:** is the cost forecast honest, is the schedule telling the truth, are the gaps between contractors going to bite.
- **In design:** which of the 3,900 late documents actually threaten the finish date.
- **At handover:** what's missing before operations will accept the asset.
- **Then the loop closes:** every finished project becomes evidence for the next approval vote. That history is the owner's own, and it compounds, which is the part no competitor can copy.